75. Let the numbers be x, $x + 2$ and $x + 4$.

Then, $x + (x + 2) + (x + 4) = x + 20$

$\Leftrightarrow$ $2x = 14$ $\Leftrightarrow$ $x = 7$.

$\therefore$ Middle number $= x + 2 = 9$.

76. Let the numbers be x, $x + 2$ and $x + 4$.

Then, $\dfrac{x(x+2)(x+4)}{8} = 720$

$\Rightarrow$ $x(x+2)(x+4) = 5760$.

$\therefore$ $\sqrt{x} \times \sqrt{(x+2)} \times \sqrt{(x+4)}$

$= \sqrt{x(x+2)(x+4)} = \sqrt{5760} = 24\sqrt{10}$.

77. Let the numbers be $3x$, $3x + 3$ and $3x + 6$.

Then, $3x + (3x + 3) + (3x + 6) = 72$

$\Leftrightarrow$ $9x = 63$

$\Leftrightarrow$ $x = 7$.

$\therefore$ Largest number $= 3x + 6 = 27$.

78. Let the numbers be x and $x + 2$.

Then, $(x + 2)^2 - x^2 = 84$

$\Leftrightarrow$ $4x + 4 = 84$

$\Leftrightarrow$ $4x = 80$

$\Leftrightarrow$ $x = 20$.

$\therefore$ Required sum $= x + (x + 2) = 2x + 2 = 42$.

79. Let the numbers be x, $x + 1$ and $x + 2$.

Then, $x^2 + (x + 1)^2 + (x + 2)^2 = 2030$

$\Leftrightarrow$ $3x^2 + 6x - 2025 = 0$

$\Leftrightarrow$ $x^2 + 2x - 675 = 0$

$\Leftrightarrow$ $(x + 27)(x - 25) = 0$

$\Leftrightarrow$ $x = 25$.

$\therefore$ Middle number $= (x + 1) = 26$.

80. $120 = 2 \times 2 \times 2 \times 3 \times 5 = (2 \times 2) \times 5 \times (2 \times 3) = 4 \times 5 \times 6$.

Clearly, the three consecutive integers whose product is 120 are 4, 5 and 6.

Required sum $= 4 + 5 + 6 = 15$.

81. Let the numbers be x and y.

Then, $2x + 3y = 39$ ...(*i*)

and $3x + 2y = 36$ ...(*ii*)

On solving (*i*) and (*ii*), we get : $x = 6$ and $y = 9$.

$\therefore$ Larger number $= 9$.

82. Let the ten's digit be x. Then, unit's digit $= 4x$.

$\therefore$ $x + 4x = 10$ $\Leftrightarrow$ $5x = 10$ $\Leftrightarrow$ $x = 2$.

So, ten's digit = 2, unit's digit = 8.

Hence, the required number is 28.

83. Let the ten's digit be x.

Then, number $= 10x + 3$ and sum of digits $= (x + 3)$.

So, $(x+3) = \dfrac{1}{7}(10x+3)$ $\Leftrightarrow$ $7x + 21$

$= 10x + 3$ $\Leftrightarrow$ $3x = 18$ $\Leftrightarrow$ $x = 6$.

Hence, the number is 63.

84. Let the ten's digit be x and the unit's digit be y.

Then, number $= 10x + y$.

$\therefore$ $10x + y = k(x + y)$ $\Rightarrow$ $k = \dfrac{10x+y}{x+y}$.

Number formed by interchanging the digits $= 10y + x$.

Let $10y + x = h(x + y)$.

Then, $h = \dfrac{10y+x}{x+y} = \dfrac{11(x+y)-(10x+y)}{x+y}$

$= 11 - \dfrac{10x+y}{x+y} = 11 - k$.

85. Let the ten's digit be x.

Then, unit's digit $= 2x$.

Number $= 10x + 2x = 12x$; Sum of digits $= x + 2x = 3x$.

$\therefore$ $12x - 3x = 18$ $\Leftrightarrow$ $9x = 18$ $\Leftrightarrow$ $x = 2$.

Hence, required number $= 12x = 24$.

86. Let the ten's digit be x and unit's digit be y.

Then, $x + y = 15$ and $x - y = 3$ or $y - x = 3$.

Solving $x + y = 15$ and $x - y = 3$, we get : $x = 9$, $y = 6$.

Solving $x + y = 15$ and $y - x = 3$, we get : $x = 6$, $y = 9$.

So, the number is either 96 or 69. Hence, the number cannot be determined.

87. Let the ten's digit be x and the unit's digit be y.

Then, number $= 10x + y$.

$\therefore$ $10x + y = 7(x + y)$ $\Leftrightarrow$ $3x = 6y$ $\Leftrightarrow$ $x = 2y$.

Number formed by reversing the digits $= 10y + x$.

$\therefore$ $(10x + y) - (10y + x) = 18 \Leftrightarrow 9x - 9y = 18 \Leftrightarrow x - y = 2 \Leftrightarrow 2y - y = 2 \Leftrightarrow y = 2$.

So, $x = 2y = 4$.

Hence, required number $= 10x + y = 40 + 2 = 42$.

88. Let the ten's digit be x and the unit's digit be y. Then, number $= 10x + y$.

New number $= 10 \times 2x + \dfrac{y}{2} = 20x + \dfrac{y}{2}$.

$\therefore$ $20x + \dfrac{y}{2} = 10y + x \Rightarrow 40x + y = 20y + 2x$

$\Rightarrow$ $38x = 19y \Rightarrow y = 2x$.

So, the unit's digit is twice the ten's digit.

89. Let the ten's digit be x. Then, unit's digit $= x + 2$.

Number $= 10x + (x + 2) = 11x + 2$; Sum of digits $= x + (x + 2) = 2x + 2$.

$\therefore$ $(11x + 2)(2x + 2) = 144$

$\Leftrightarrow$ $22x^2 + 26x - 140 = 0$

$\Leftrightarrow$ $11x^2 + 13x - 70 = 0$

$\Leftrightarrow$ $(x - 2)(11x + 35) = 0$

$\Leftrightarrow$ $x = 2$.

Hence, required number $= 11x + 2 = 24$.

90. Let the ten's digit be x and unit's digit be y.

Then, number $= 10x + y$.

Number obtained by interchanging the digits $= 10y + x$.

$\therefore$ $(10x + y) + (10y + x)$

$= 11(x + y)$, which is divisible by 11.

91. Let the ten's digit be x and unit's digit be y.

Then, $(10x + y) - (x + y) = 9$ or $x = 1$.

From this data, we cannot find y, the unit's digit.

So, the data is inadequate.